

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399699
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Radhakrishnan; Vineeth et al.

Systems and methods for performing health score-based firmware updates

Abstract

Embodiments of the present disclosure provide a firmware update to multiple, similar hardware devices in an Information Handling System (IHS). In an illustrative, non-limiting embodiment, an IHS may include computer-executable instructions to receive a request to perform a firmware update to a plurality of devices, update a first of the plurality of devices with the requested firmware update, and calculate a health score for the first device as a result of the first device being updated. Based upon the health score, the system and method may determine whether to update the other devices based at least in part, on the determined health score, and update the other devices based on the determination.

Inventors: Radhakrishnan; Vineeth (Palakkad, IN), Ramaiah; Dharma Bhushan (Bangalore, IN), Thankappan; Mini Thottunkal (Bangalore, IN), Bisa; Rama Rao (Bangalore, IN), Rahiman; Shinose Abdul (Bangalore, IN)

Applicant: Dell Products, L.P. (Round Rock, TX)

Family ID: 1000008781635

Assignee: Dell Products, L.P. (Round Rock, TX)

Appl. No.: 17/934653

Filed: September 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240103825 A1	Mar. 28, 2024

Publication Classification

Int. Cl.: G06F8/65 (20180101)

U.S. Cl.:

CPC **G06F8/65** (20130101);

Field of Classification Search

CPC: G06F (8/65); G06F (8/654)

USPC: 717/168

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10409584	12/2018	Kulchytskyy et al.	N/A	G06F 8/65
10642623	12/2019	Righi et al.	N/A	G06F 8/65
11093321	12/2020	Zeavelou et al.	N/A	G06F 8/65
2015/0160627	12/2014	Maddukuri et al.	700/90	G06F 8/65
2016/0036956	12/2015	Debates et al.	455/419	G06F 8/65
2020/0019393	12/2019	Vichare	N/A	G06N 20/00
2020/0034133	12/2019	Dattatri	N/A	G06F 8/65
2020/0372157	12/2019	Singer et al.	N/A	G06F 8/64
2020/0387390	12/2019	Bat Ulzii	N/A	G06F 40/274
2023/0017316	12/2022	Kulkarni	N/A	G06N 5/04

Primary Examiner: Chen; Qing

Attorney, Agent or Firm: Fogarty LLP

Background/Summary

BACKGROUND

(1) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is Information Handling Systems (IHSs). An IHS generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, IHSs may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in IHSs allow for IHSs to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, IHSs may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

(2) Various hardware components of an IHS may operate using firmware instructions. From time to time, it is expected that firmware utilized by hardware components of an IHS may be updated. Such firmware updates may be made in order to modify the capabilities of a particular hardware

component, such as to address security vulnerabilities or to adapt the operations of the hardware component to a specific computing task. When firmware updates are made to a hardware component of an IHS, it is preferable that the IHS experience no downtime and with minimal degradation in the performance of the IHS.

(3) Nowadays, software updates are typically made available on one or more download sites as soon as the software provider can produce them. In this manner, software providers can be more responsive to critical flaws, security concerns, and general customer needs. As a result, to update software, a customer would query an update site for software updates, and download and install the software update if available. For example, a typical network-based software update procedure may include the steps of issuing a request over a network to a software provider's download site (e.g., update source) for a software update applicable to the client computer. The update source responds to the client computer with the software update requested by the client computer in the update request. After the client computer has received the software update, the client computer installs the received software update.

(4) One benefit of updating software in such a manner is the reduced cost associated with producing and distributing software updates. Additionally, software updates can now be performed more frequently, especially those that address critical issues and security. Still further, a computer user has greater control as to when and which software updates should be installed on the client computer.

SUMMARY

(5) Embodiments of the present disclosure provide a firmware update to multiple, similar hardware devices in an Information Handling System (IHS). In an illustrative, non-limiting embodiment, an IHS may include computer-executable instructions to receive a request to perform a firmware update to a plurality of devices, update a first of the plurality of devices with the requested firmware update, and calculate a health score for the first device as a result of the first device being updated. Based upon the health score, the system and method may determine whether to update the other devices based at least in part, on the determined health score, and update the other devices based on the determination.

(6) According to another embodiment, a health score-based firmware updating method includes the steps of receiving a request to perform a firmware update to a plurality of devices, updating a first of the plurality of devices with the requested firmware update, and calculating a health score for the first device as a result of the first device being updated. Based on the determined health score, the method further includes the steps of determining whether to update the other devices based at least in part, on the determined health score, and updating the other devices based on the determination.

(7) According to yet another embodiment, a memory storage device is configured with program instructions that, upon execution by a client Information Handling System (IHS), cause the client IHS to receive a request to perform a firmware update to a plurality of devices, update a first of the plurality of devices with the requested firmware update, and calculate a health score for the first device as a result of the first device being updated. Based on the determined health score, the instructions further cause the IHS to determine whether to update the other devices based at least in part, on the determined health score, and update the other devices based on the determination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention(s) is/are illustrated by way of example and is/are not limited by the accompanying figures. Elements in the figures are illustrated for simplicity and clarity, and have not necessarily been drawn to scale.

(2) FIGS. 1A and 1B illustrate certain components of a chassis comprising one or more compute

sleds and one or more storage sleds that may be configured to implement the systems and methods described according to one embodiment of the present disclosure.

(3) FIG. 2 illustrates an example IHS configured to implement systems and methods described herein according to one embodiment of the present disclosure.

(4) FIG. 3 is a diagram illustrating several components of an example associated IHS showing how those components may signal one another for implementing the health score-based firmware updating system and method according to one embodiment of the present disclosure.

(5) FIGS. 4A and 4B illustrate a health score-based firmware updating method depicting how the hardware devices of the IHS may receive a firmware update according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

(6) The present disclosure is described with reference to the attached figures. The figures are not drawn to scale, and they are provided merely to illustrate the disclosure. Several aspects of the disclosure are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide an understanding of the disclosure. The present disclosure is not limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are required to implement a methodology in accordance with the present disclosure.

(7) Customers often upgrade the firmware in the IHSs of a data center for assorted reasons, such as to meet compliance policies, to take advantage of new features, enhancements, deploy security fixes, and the like. The IHSs that are NVMe-MI/PLDM Specification compliant can take advantage of updating firmware in parallel to all IHSs in a system or in a cluster without rebooting the servers. The devices which support Platform Level Data Model (PLDM) offers an option for the Remote Access Controller (RAC) to update the firmware without rebooting the HOST server. Thus, downtime is often not incurred during the firmware update process.

(8) In a cluster environment or other system where multiple PLDM compliant devices (IHSs) belonging to same vendor exist, there is a possibility that a firmware update may be triggered for all the devices in an IHS concurrently. The firmware update may, in some cases, cause certain issues that were not observed during development of the firmware update (e.g., in a test environment). Examples of such issues may include a different run time environment (e.g., mismatch between in-band driver and side-band firmware), configuration-based issues, side band issues, etc.). These unwanted issues can damage or bring down the cluster or system on which the firmware update was performed. As will be described in detail herein below, embodiments of the present disclosure provide a health score-based firmware updating system and method that performs a firmware update on a single IHS, and calculates a health score for that IHS. Using the calculated health score, either the user (cluster administrator) or system may determine whether to perform the firmware update on the other IHSs in the cluster.

(9) FIGS. 1A and 1B illustrate certain components of a chassis **100** comprising one or more compute sleds **105a-n** and one or more storage sleds **115a-n** that may be configured to implement the systems and methods described according to one embodiment of the present disclosure. Embodiments of chassis **100** may include a wide variety of hardware configurations in which one or more sleds **105a-n**, **115a-n** are installed in chassis **100**. Such variations in hardware configuration may result from chassis **100** being factory assembled to include components specified by a customer that has contracted for manufacture and delivery of chassis **100**. Upon delivery and deployment of a chassis **100**, the chassis **100** may be modified by replacing and/or adding various hardware components, in addition to replacement of the removable sleds **105a-n**, **115a-n** that are installed in the chassis. In addition, once the chassis **100** has been deployed, firmware used by individual hardware components of the sleds **105a-n**, **115a-n**, or by other hardware components of chassis **100**, may be modified in order to update the operations that are

supported by these hardware components.

(10) Chassis **100** may include one or more bays that each receive an individual sled (that may be additionally or alternatively referred to as a tray, blade, and/or node), such as compute sleds **105a-n** and storage sleds **115a-n**. Chassis **100** may support a variety of different numbers (e.g., 4, 8, 16, 32), sizes (e.g., single-width, double-width) and physical configurations of bays. Embodiments may include additional types of sleds that provide various storage, power and/or processing capabilities. For instance, sleds installable in chassis **100** may be dedicated to providing power management or networking functions. Sleds may be individually installed and removed from the chassis **100**, thus allowing the computing and storage capabilities of a chassis to be reconfigured by swapping the sleds with diverse types of sleds, in some cases at runtime without disrupting the ongoing operations of the other sleds installed in the chassis **100**.

(11) Multiple chassis **100** may be housed within a rack. Data centers may utilize large numbers of racks, with various different types of chassis installed in various configurations of racks. The modular architecture provided by the sleds, chassis and racks allow for certain resources, such as cooling, power, and network bandwidth, to be shared by the compute sleds **105a-n** and storage sleds **115a-n**, thus providing efficiency improvements and supporting greater computational loads. For instance, certain computational tasks, such as computations used in machine learning and other artificial intelligence systems, may utilize computational and/or storage resources that are shared within an IHS, within an individual chassis **100** and/or within a set of IHSs that may be spread across multiple chassis of a data center.

(12) Implementing computing systems that span multiple processing components of chassis **100** is aided by high-speed data links between these processing components, such as PCIe connections that form one or more distinct PCIe switch fabrics that are implemented by PCIe switches **135a-n**, **165a-n** installed in the sleds **105a-n**, **115a-n** of the chassis. These high-speed data links may be used to support algorithm implementations that span multiple processing, networking, and storage components of an IHS and/or chassis **100**. For instance, computational tasks may be delegated to a specific processing component of an IHS, such as to a hardware accelerator **185a-n** that may include one or more programmable processors that operate separate from the main CPUs **170a-n** of computing sleds **105a-n**. In various embodiments, such hardware accelerators **185a-n** may include DPUs (Data Processing Units), GPUs (Graphics Processing Units), SmartNICs (Smart Network Interface Card) and/or FPGAs (Field Programmable Gate Arrays). These hardware accelerators **185a-n** operate according to firmware instructions that may be occasionally updated, such as to adapt the capabilities of the respective hardware accelerators **185a-n** to specific computing tasks.

(13) Chassis **100** may be installed within a rack structure that provides at least a portion of the cooling utilized by the sleds **105a-n**, **115a-n** installed in chassis **100**. In supporting airflow cooling, a rack may include one or more banks of cooling fans that may be operated to ventilate heated air from within the chassis **100** that is housed within the rack. The chassis **100** may alternatively or additionally include one or more cooling fans **130** that may be similarly operated to ventilate heated air away from sleds **105a-n**, **115a-n** installed within the chassis. In this manner, a rack and a chassis **100** installed within the rack may utilize various configurations and combinations of cooling fans to cool the sleds **105a-n**, **115a-n** and other components housed within chassis **100**.

(14) The sleds **105a-n**, **115a-n** may be individually coupled to chassis **100** via connectors that correspond to the bays provided by the chassis **100** and that physically and electrically couple an individual sled to a backplane **160**. Chassis backplane **160** may be a printed circuit board that includes electrical traces and connectors that are configured to route signals between the various components of chassis **100** that are connected to the backplane **160** and between different components mounted on the printed circuit board of the backplane **160**. In the illustrated embodiment, the connectors for use in coupling sleds **105a-n**, **115a-n** to backplane **160** may include PCIe couplings that support high-speed data links with the sleds **105a-n**, **115a-n**. In various embodiments, backplane **160** may support diverse types of connections, such as cables, wires,

midplanes, connectors, expansion slots, and multiplexers. In certain embodiments, backplane **160** may be a motherboard that includes various electronic components installed thereon. Such components installed on a motherboard backplane **160** may include components that implement all or part of the functions described with regard to the SAS (Serial Attached SCSI) expander **150**, I/O controllers **145**, network controller **140**, chassis management controller **125** and power supply unit **135**.

(15) In certain embodiments, each individual sled **105a-n**, **115a-n-n** may be an IHS such as described with regard to IHS **200** of FIG. 2. Sleds **105a-n**, **115a-n** may individually or collectively provide computational processing resources that may be used to support a variety of e-commerce, multimedia, business, and scientific computing applications, such as artificial intelligence systems provided via cloud computing implementations. Sleds **105a-n**, **115a-n** are typically configured with hardware and software that provide leading-edge computational capabilities. Accordingly, services that are provided using such computing capabilities are typically provided as high-availability systems that operate with minimum downtime.

(16) In high-availability computing systems, such as may be implemented using embodiments of chassis **100**, any downtime that can be avoided is preferred. As described above, firmware updates are expected in the administration and operation of data centers, but it is preferable to avoid any downtime in making such firmware updates. For instance, in updating the firmware of the individual hardware components of the chassis **100**, it is preferable that such updates can be made without having to reboot the chassis. As described in additional detail below, it is also preferable that updates to the firmware of individual hardware components of sleds **105a-n**, **115a-n** be likewise made without having to reboot the respective sled of the hardware component that is being updated.

(17) As illustrated, each sled **105a-n**, **115a-n** includes a respective remote access controller (RAC) **110a-n**, **120a-n**. As described in additional detail with regard to FIG. 2, remote access controller **110a-n**, **120a-n** provides capabilities for remote monitoring and management of a respective sled **105a-n**, **115a-n** and/or of chassis **100**. In support of these monitoring and management functions, remote access controllers **110a-n** may utilize both in-band and side-band (i.e., out-of-band) communications with various managed components of a respective sled **105a-n** and chassis **100**. Remote access controllers **110a-n**, **120a-n** may collect diverse types of sensor data, such as collecting temperature sensor readings that are used in support of airflow cooling of the chassis **100** and the sleds **105a-n**, **115a-n**. In addition, each remote access controller **110a-n**, **120a-n** may implement various monitoring and administrative functions related to a respective sleds **105a-n**, **115a-n**, where these functions may be implemented using sideband bus connections with various internal components of the chassis **100** and of the respective sled **105a-n**, **115a-n**. As described in additional detail below, in various embodiments, these capabilities of the remote access controllers **110a-n**, **120a-n** may be utilized in updating the firmware of hardware components of chassis **100** and/or of hardware components of the sleds **105a-n**, **115a-n**, without having to reboot the chassis or any of the sleds **105a-n**, **115a-n**.

(18) The remote access controllers **110a-n**, **120a-n** that are present in chassis **100** may support secure connections with a remote management interface **101**. In some embodiments, remote management interface **101** provides a remote administrator with various capabilities for remotely administering the operation of an IHS, including initiating updates to the firmware used by hardware components installed in the chassis **100**. For example, remote management interface **101** may provide capabilities by which an administrator can initiate updates to all of the storage drives **175a-n** installed in a chassis **100**, or to all of the storage drives **175a-n** of a particular model or manufacturer. In some instances, remote management interface **101** may include an inventory of the hardware, software, and firmware of chassis **100** that is being remotely managed through the operation of the remote access controllers **110a-n**, **120a-n**. The remote management interface **101** may also include various monitoring interfaces for evaluating telemetry data collected by the

remote access controllers **110a-n**, **120a-n**. In some embodiments, remote management interface **101** may communicate with remote access controllers **110a-n**, **120a-n** via a protocol such the Redfish remote management interface.

(19) In the illustrated embodiment, chassis **100** includes one or more compute sleds **105a-n** that are coupled to the backplane **160** and installed within one or more bays or slots of chassis **100**. Each of the individual compute sleds **105a-n** may be an IHS, such as described with regard to FIG. 2. Each of the individual compute sleds **105a-n** may include various different numbers and types of processors that may be adapted to performing specific computing tasks. In the illustrated embodiment, each of the compute sleds **105a-n** includes a PCIe switch **135a-n** that provides access to a hardware accelerator **185a-n**, such as the described DPUs, GPUs, Smart NICs and FPGAs, which may be programmed and adapted for specific computing tasks, such as to support machine learning or other artificial intelligence systems. As described in additional detail below, compute sleds **105a-n** may include a variety of hardware components, such as hardware accelerator **185a-n** and PCIe switches **135a-n**, that operate using firmware that may be occasionally updated.

(20) As illustrated, chassis **100** includes one or more storage sleds **115a-n** that are coupled to the backplane **160** and installed within one or more bays of chassis **100** in a similar manner to compute sleds **105a-n**. Each of the individual storage sleds **115a-n** may include various different numbers and types of storage devices. As described in additional detail with regard to FIG. 2, a storage sled **115a-n** may be an IHS **200** that includes multiple solid-state drives (SSDs) **175a-n**, where the individual storage drives **175a-n** may be accessed through a PCIe switch **165a-n** of the respective storage sled **115a-n**.

(21) As illustrated, a storage sled **115a** may include one or more DPUs (Data Processing Units) **190** that provide access to and manage the operations of the storage drives **175a** of the storage sled **115a**. Use of a DPU **190** in this manner provides low-latency and high-bandwidth access to numerous storage drives **175a**. These storage drives **175a** may be utilized in parallel through NVMe transmissions that are supported by the PCIe switch **165a** that connects the SSDs **175a** to the DPU **190**. In some instances, PCIe switch **165a** may be an integrated component of a DPU **190**. The immense data storage and retrieval capabilities provided by such storage sled **115a** implementations may be harnessed by offloading storage operations directed as storage drives **175a** to a DPU **190**, and thus without relying on the main CPU of the storage sled, or of any other component of chassis **100**. As indicated in FIG. 1, chassis **100** may also include one or more storage sleds **115n** that provide access to storage drives **175n** via a storage controller **195**. In some embodiments, storage controller **195** may provide support for RAID (Redundant Array of Independent Disks) configurations of logical and physical storage drives, such as storage drives provided by storage sled **115n**. In some embodiments, storage controller **195** may be a HBA (Host Bus Adapter) that provides more limited capabilities in accessing storage drives **175n**.

(22) In addition to the data storage capabilities provided by storage sleds **115a-n**, chassis **100** may provide access to other storage resources that may be installed components of chassis **100** and/or may be installed elsewhere within a rack that houses the chassis **100**. In certain scenarios, such storage drives **155** may be accessed via a SAS expander **150** that is coupled to the backplane **160** of the chassis **100**. The SAS expander **150** may support connections to a number of JBOD (Just a Bunch of Disks) storage drives **155** that, in some instances, may be configured and managed individually and without implementing data redundancy across the various storage drives **155**. The additional storage drives **155** may also be at various other locations within a datacenter in which chassis **100** is installed.

(23) In light of the various manners in which storage drives **175a-n**, **155** may be coupled to chassis **100**, a wide variety of different storage topologies may be supported. Through these supported topologies, storage drives **175a-n**, **155** may be logically organized into clusters or other groupings that may be collectively tasked and managed. In some instances, a chassis **100** may include numerous storage drives **175a-n**, **155** that are identical, or nearly identical, such as arrays of storage

drives of the same manufacturer and model. Accordingly, any firmware updates to storage drives **175a-n**, **155** require the updates to be applied within each of these topologies being supported by the chassis **100**. Despite the substantial number of different storage drive topologies that may be supported by an individual chassis **100**, the firmware used by each of these storage devices **175a-n**, **155** may be occasionally updated. In some instances, firmware updates may be limited to a single storage drive, but in other instance, firmware updates may be initiated for a large number of storage drives, such as for all SSDs installed in chassis **100**.

(24) As illustrated, the chassis **100** of FIG. **1** includes a network controller **140** that provides network access to the sleds **105a-n**, **115a-n** installed within the chassis. Network controller **140** may include various switches, adapters, controllers, and couplings used to connect chassis **100** to a network, either directly or via additional networking components and connections provided via a rack in which chassis **100** is installed. Network controller **140** operates according to firmware instructions that may be occasionally updated.

(25) Chassis **100** may similarly include a power supply unit **135** that provides the components of the chassis with various levels of DC power from an AC power source or from power delivered via a power system provided by a rack within which chassis **100** may be installed. In certain embodiments, power supply unit **135** may be implemented within a sled that may provide chassis **100** with redundant, hot-swappable power supply units. Power supply unit **135** may operate according to firmware instructions that may be occasionally updated.

(26) Chassis **100** may also include various I/O controllers **145** that may support various I/O ports, such as USB ports that may be used to support keyboard and mouse inputs and/or video display capabilities. Each of the I/O controllers **145** may operate according to firmware instructions that may be occasionally updated. Such I/O controllers **145** may be utilized by the chassis management controller **125** to support various KVM (Keyboard, Video and Mouse) **125a** capabilities that provide administrators with the ability to interface with the chassis **100**. The chassis management controller **125** may also include a storage module **125c** that provides capabilities for managing and configuring certain aspects of the storage devices of chassis **100**, such as the storage devices provided within storage sleds **115a-n** and within the storage drives **155** configured in a JBOD.

(27) In addition to providing support for KVM **125a** capabilities for administering chassis **100**, chassis management controller **125** may support various additional functions for sharing the infrastructure resources of chassis **100**. In some scenarios, chassis management controller **125** may implement tools for managing the power supply unit **135**, network controller **140** and airflow cooling fans **130** that are available via the chassis **100**. As described, the cooling fans **130** utilized by chassis **100** may include an airflow cooling system that is provided by a rack in which the chassis **100** may be installed and managed by a cooling module **125b** of the chassis management controller **125**.

(28) For purposes of this disclosure, an IHS may include any instrumentality or aggregate of instrumentalities operable to compute, calculate, determine, classify, process, transmit, receive, retrieve, originate, switch, store, display, communicate, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an IHS may be a personal computer (e.g., desktop or laptop), tablet computer, mobile device (e.g., Personal Digital Assistant (PDA) or smart phone), server (e.g., blade server or rack server), a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. An IHS may include Random Access Memory (RAM), one or more processing resources such as a Central Processing Unit (CPU) or hardware or software control logic, Read-Only Memory (ROM), and/or other types of nonvolatile memory. Additional components of an IHS may include one or more disk drives, one or more network ports for communicating with external devices as well as various I/O devices, such as a keyboard, a mouse, touchscreen, and/or a video display. As described, an IHS may also include one or more buses operable to transmit communications between the various hardware components. An example

of an IHS is described in more detail below.

(29) FIG. 2 illustrates an example of an IHS **200** configured to implement systems and methods described herein according to one embodiment of the present disclosure. It should be appreciated that although the embodiments described herein may describe an IHS that is a compute sled or similar computing component that may be deployed within the bays of a chassis, a variety of other types of IHSs, such as laptops and portable devices, may also operate according to embodiments described herein. In the illustrative embodiment of FIG. 2, IHS **200** may be a computing component, such as sled **105a-n**, **115a-n**, or other type of server, such as an 1 RU server installed within a 2 RU chassis, which is configured to share infrastructure resources provided within a chassis **100**.

(30) IHS **200** may utilize one or more system processors **205**, that may be referred to as CPUs (central processing units). In some embodiments, CPUs **205** may each include a plurality of processing cores that may be separately delegated with computing tasks. Each of the CPUs **205** may be individually designated as a main processor and as a co-processor, where such designations may be based on delegation of specific types of computational tasks to a CPU **205**. In some embodiments, CPUs **205** may each include an integrated memory controller that may be implemented directly within the circuitry of each CPU **205**. In some embodiments, a memory controller may be a separate integrated circuit that is located on the same die as the CPU **205**. Each memory controller may be configured to manage the transfer of data to and from a system memory **210** of the IHS, in some cases using a high-speed memory bus **205a**. The system memory **210** is coupled to CPUs **205** via one or more memory buses **205a** that provide the CPUs **205** with high-speed memory used in the execution of computer program instructions by the CPUs **205**. Accordingly, system memory **210** may include memory components, such as static RAM (SRAM), dynamic RAM (DRAM), NAND Flash memory, suitable for supporting high-speed memory operations by the CPUs **205**. In certain embodiments, system memory **210** may combine persistent non-volatile memory and volatile memory.

(31) In certain embodiments, the system memory **210** may be comprised of multiple removable memory modules. The system memory **210** of the illustrated embodiment includes removable memory modules **210a-n**. Each of the removable memory modules **210a-n** may correspond to a printed circuit board memory socket that receives a removable memory module **210a-n**, such as a DIMM (Dual In-line Memory Module), that can be coupled to the socket and then decoupled from the socket as needed, such as to upgrade memory capabilities or to replace faulty memory modules. Other embodiments of IHS system memory **210** may be configured with memory socket interfaces that correspond to diverse types of removable memory module form factors, such as a Dual In-line Package (DIP) memory, a Single In-line Pin Package (SIPP) memory, a Single In-line Memory Module (SIMM), and/or a Ball Grid Array (BGA) memory.

(32) IHS **200** may utilize a chipset that may be implemented by integrated circuits that are connected to each CPU **205**. All or portions of the chipset may be implemented directly within the integrated circuitry of an individual CPU **205**. The chipset may provide the CPU **205** with access to a variety of resources accessible via one or more in-band buses. IHS **200** may also include one or more I/O ports **215** that may be used to couple the IHS **200** directly to other IHSs, storage resources, diagnostic tools, and/or other peripheral components. A variety of additional components may be coupled to CPUs **205** via a variety of in-line buses. For instance, CPUs **205** may also be coupled to a power management unit **220** that may interface with a power system of the chassis **100** in which IHS **200** may be installed. In addition, CPUs **205** may collect information from one or more sensors **225** via a management bus.

(33) In certain embodiments, IHS **200** may operate using a BIOS (Basic Input/Output System) that may be stored in a non-volatile memory accessible by the CPUs **205**. The BIOS may provide an abstraction layer by which the operating system of the IHS **200** interfaces with hardware components of the IHS. Upon powering or restarting IHS **200**, CPUs **205** may utilize BIOS

instructions to initialize and test hardware components coupled to the IHS, including both components permanently installed as components of the motherboard of IHS **200** and removable components installed within various expansion slots supported by the IHS **200**. The BIOS instructions may also load an operating system for execution by CPUs **205**. In certain embodiments, IHS **200** may utilize Unified Extensible Firmware Interface (UEFI) in addition to or instead of a BIOS. In certain embodiments, the functions provided by a BIOS may be implemented, in full or in part, by the remote access controller **230**.

(34) In some embodiments, IHS **200** may include a TPM (Trusted Platform Module) that may include various registers, such as platform configuration registers, and a secure storage, such as an NVRAM (Non-Volatile Random-Access Memory). The TPM may also include a cryptographic processor that supports various cryptographic capabilities. In IHS embodiments that include a TPM, a pre-boot process implemented by the TPM may utilize its cryptographic capabilities to calculate hash values that are based on software and/or firmware instructions utilized by certain core components of IHS, such as the BIOS and boot loader of IHS **200**. These calculated hash values may then be compared against reference hash values that were previously stored in a secure non-volatile memory of the IHS, such as during factory provisioning of IHS **200**. In this manner, a TPM may establish a root of trust that includes core components of IHS **200** that are validated as operating using instructions that originate from a trusted source.

(35) As illustrated, CPUs **205** may be coupled to a network controller **240**, such as provided by a Network Interface Controller (NIC) card that provides IHS **200** with communications via one or more external networks, such as the Internet, a LAN, or a WAN. In some embodiments, network controller **240** may be a replaceable expansion card or adapter that is coupled to a connector (e.g., PCIe connector of a motherboard, backplane, midplane, etc.) of IHS **200**. In some embodiments, network controller **240** may support high-bandwidth network operations by the IHS **200** through a PCIe interface that is supported by the chipset of CPUs **205**. Network controller **240** may operate according to firmware instructions that may be occasionally updated.

(36) As indicated in FIG. 2, in some embodiments, CPUs **205** may be coupled to a PCIe card **255** that includes two PCIe switches **265a-b** that operate as I/O controllers for PCIe communications, such as TLPs (Transaction Layer Packets), that are transmitted between the CPUs **205** and PCIe devices and systems coupled to IHS **200**. Whereas the illustrated embodiment of FIG. 2 includes two CPUs **205** and two PCIe switches **265a-b**, different embodiments may operate using different numbers of CPUs and PCIe switches. In addition to serving as I/O controllers that route PCIe traffic, PCIe switches **265a-b** include switching logic that can be used to expand the number of PCIe connections that are supported by CPUs **205**. PCIe switches **265a-b** may multiply the number of PCIe lanes available to CPUs **205**, thus allowing more PCIe devices to be connected to CPUs **205**, and for the available PCIe bandwidth to be allocated with greater granularity. Each of the PCIe switches **265a-b** may operate according to firmware instructions that may be occasionally updated.

(37) Using the available PCIe lanes, the PCIe switches **265a-b** may be used to implement a PCIe switch fabric. Also through this switch fabric, PCIe NVMe (Non-Volatile Memory Express) transmission may be supported and utilized in high-speed communications with SSDs, such as storage drives **235a-b**, of the IHS **200**. Also through this switch fabric, PCIe VDM (Vendor Defined Messaging) may be supported and utilized in managing PCIe-compliant hardware components of the IHS **200**, such as in updating the firmware utilized by the hardware components.

(38) As indicated in FIG. 2, IHS **200** may support storage drives **235a-b** in various topologies, in the same manner as described with regard to the chassis **100** of FIG. 1. In the illustrated embodiment, storage drives **235a** are accessed via a hardware accelerator **250**, while storage drives **235b** are accessed directly via PCIe switch **265b**. In some embodiments, the storage drives **235a-b** of IHS **200** may include a combination of both SSD and magnetic disk storage drives. In other embodiments, all of the storage drives **235a-b** of IHS **200** may be identical, or nearly identical. In all embodiments, storage drives **235a-b** operate according to firmware instructions that may be

occasionally updated.

(39) As illustrated, PCIe switch **265a** is coupled via a PCIe link to a hardware accelerator **250**, such as a DPU, SmartNIC, GPU and/or FPGA, that may be connected to the IHS via a removable card or baseboard that couples to a PCIe connector of the IHS **200**. In some embodiments, hardware accelerator **250** includes a programmable processor that can be configured for offloading functions from CPUs **205**. In some embodiments, hardware accelerator **250** may include a plurality of programmable processing cores and/or hardware accelerators, which may be used to implement functions used to support devices coupled to the IHS **200**. In some embodiments, the processing cores of hardware accelerator **250** include ARM (advanced RISC (reduced instruction set computing) machine) processing cores. In other embodiments, the cores of the DPUs may include MIPS (microprocessor without interlocked pipeline stages) cores, RISC-V cores, or CISC (complex instruction set computing) (i.e., x86) cores. Hardware accelerator **250** may operate according to firmware instructions that may be occasionally updated.

(40) In the illustrated embodiment, the programmable capabilities of hardware accelerator **250** implement functions used to support storage drives **235a**, such as SSDs. In such storage drive topologies, hardware accelerator **250** may implement processing of PCIe NVMe communications with SSDs **235a**, thus supporting high-bandwidth connections with these SSDs. Hardware accelerator **250** may also include one or more memory devices used to store program instructions executed by the processing cores and/or used to support the operation of SSDs **235a** such as in implementing cache memories and buffers utilized in support of high-speed operation of these storage drives, and in some cases may be used to provide high-availability and high-throughput implementations of the read, write and other I/O operations that are supported by these storage drives **235a**. In other embodiments, hardware accelerator **250** may implement operations in support of other types of devices and may similarly support high-bandwidth PCIe connections with these devices. For instance, in various embodiments, hardware accelerator **250** may support high-bandwidth connections, such as PCIe connections, with networking devices in implementing functions of a network switch, compression and codec functions, virtualization operations or cryptographic functions.

(41) As illustrated in FIG. 2, PCIe switches **265a-b** may also support PCIe couplings with one or more GPUs (Graphics Processing Units) **260**. Embodiments may include one or more GPU cards, where each GPU card is coupled to one or more of the PCIe switches **265a-b**, and where each GPU card may include one or more GPUs **260**. In some embodiments, PCIe switches **265a-b** may transfer instructions and data for generating video images by the GPUs **260** to and from CPUs **205**. Accordingly, GPUs **260** may include one or more hardware-accelerated processing cores that are optimized for performing streaming calculation of vector data, matrix data and/or other graphics data, thus supporting the rendering of graphics for display on devices coupled either directly or indirectly to IHS **200**. In some instances, GPUs **260** may be utilized as programmable computing resources for offloading other functions from CPUs **205**, in the same manner as hardware accelerator **250**. GPUs **260** may operate according to firmware instructions that may be occasionally updated.

(42) As illustrated in FIG. 2, PCIe switches **265a-b** may support PCIe connections in addition to those utilized by GPUs **260** and hardware accelerator **250**, where these connections may include PCIe links of one or more lanes. For instance, PCIe connectors **245** supported by a printed circuit board of IHS **200** may allow various other systems and devices to be coupled to IHS. Through couplings to PCIe connectors **245**, a variety of data storage devices, graphics processors and network interface cards may be coupled to IHS **200**, thus supporting a wide variety of topologies of devices that may be coupled to the IHS **200**.

(43) As described, IHS **200** includes a remote access controller **230** that supports remote management of IHS **200** and of various internal components of IHS **200**. In certain embodiments, remote access controller **230** may operate from a different power plane from the processors **205**

and other components of IHS **200**, thus allowing the remote access controller **230** to operate, and management tasks to proceed, while the processing cores of IHS **200** are powered off. Various functions provided by the BIOS, including launching the operating system of the IHS **200**, and/or functions of a TPM may be implemented or supplemented by the remote access controller **230**. In some embodiments, the remote access controller **230** may perform various functions to verify the integrity of the IHS **200** and its hardware components prior to initialization of the operating system of IHS **200** (i.e., in a bare-metal state). In some embodiments, certain operations of the remote access controller **230**, such as the operations described herein for updating firmware used by managed hardware components of IHS **200**, may operate using validated instructions, and thus within the root of trust of IHS **200**.

(44) In some embodiments, remote access controller **230** may include a service processor **230a**, or specialized microcontroller, which operates management software that supports remote monitoring and administration of IHS **200**. The management operations supported by remote access controller **230** may be remotely initiated, updated, and monitored via a remote management interface **101**, such as described with regard to FIG. **1**. Remote access controller **230** may be installed on the motherboard of IHS **200** or may be coupled to IHS **200** via an expansion slot or other connector provided by the motherboard. In some instances, the management functions of the remote access controller **230** may utilize information collected by various managed sensors **225** located within the IHS. For instance, temperature data collected by sensors **225** may be utilized by the remote access controller **230** in support of closed-loop airflow cooling of the IHS **200**. As indicated, remote access controller **230** may include a secured memory **230e** for exclusive use by the remote access controller in support of management operations.

(45) In some embodiments, remote access controller **230** may implement monitoring and management operations using MCTP (Management Component Transport Protocol) messages that may be communicated to managed devices **205**, **235a-b**, **240**, **250**, **255**, **260** via management connections supported by a sideband bus **253**. In some embodiments, the remote access controller **230** may additionally or alternatively use MCTP messaging to transmit Vendor Defined Messages (VDMs) via the in-line PCIe switch fabric supported by PCIe switches **265a-b**. In some instances, the sideband management connections supported by remote access controller **230** may include PLDM (Platform Level Data Model) management communications with the managed devices **205**, **235a-b**, **240**, **250**, **255**, **260** of IHS **200**.

(46) As illustrated, remote access controller **230** may include a network adapter **230c** that provides the remote access controller with network access that is separate from the network controller **240** utilized by other hardware components of the IHS **200**. Through secure connections supported by network adapter **230c**, remote access controller **230** communicates management information with remote management interface **101**. In support of remote monitoring functions, network adapter **230c** may support connections between remote access controller **230** and external management tools using wired and/or wireless network connections that operate using a variety of network technologies. As a non-limiting example of a remote access controller, the integrated Dell Remote Access Controller (iDRAC) from Dell® is embedded within Dell servers and provides functionality that helps information technology (IT) administrators deploy, update, monitor, and maintain servers remotely.

(47) Remote access controller **230** supports monitoring and administration of the managed devices of an IHS via a sideband bus interface **255**. For instance, messages utilized in device and/or system management may be transmitted using I2C sideband bus **253** connections that may be individually established with each of the respective managed devices **205**, **235a-b**, **240**, **250**, **255**, **260** of the IHS **200** through the operation of an I2C multiplexer **230d** of the remote access controller. As illustrated in FIG. **2**, the managed devices **205**, **235a-b**, **240**, **250**, **255**, **260** of IHS **200** are coupled to the CPUs **205**, either directly or indirectly, via in-line buses that are separate from the I2C sideband bus **253** connections used by the remote access controller **230** for device management.

(48) In certain embodiments, the service processor **230a** of remote access controller **230** may rely on an I2C co-processor **230b** to implement sideband I2C communications between the remote access controller **230** and the managed hardware components **205**, **235a-b**, **240**, **250**, **255**, **260** of the IHS **200**. The I2C co-processor **230b** may be a specialized co-processor or micro-controller that is configured to implement a I2C bus interface used to support communications with managed hardware components **205**, **235a-b**, **240**, **250**, **255**, **260** of IHS. In some embodiments, the I2C co-processor **230b** may be an integrated circuit on the same die as the service processor **230a**, such as a peripheral system-on-chip feature that may be provided by the service processor **230a**. The sideband I2C bus **255** is illustrated as single line in FIG. 2. However, sideband bus **255** may be comprised of multiple signaling pathways, where each may be comprised of a clock line and data line that couple the remote access controller **230** to I2C endpoints **205**, **235a-b**, **240**, **250**, **255**, **260**.

(49) In various embodiments, an IHS **200** does not include each of the components shown in FIG. 2. In various embodiments, an IHS **200** may include various additional components in addition to those that are shown in FIG. 2. Furthermore, some components that are represented as separate components in FIG. 2 may in certain embodiments instead be integrated with other components. For example, in certain embodiments, all or a portion of the functionality provided by the illustrated components may instead be provided by components integrated into the one or more processor(s) **205** as a systems-on-a-chip.

(50) FIG. 3 is a diagram **300** illustrating several components of an example associated IHS **200** showing how those components may signal one another for implementing the health score-based firmware updating system and method according to one embodiment of the present disclosure. The IHS **200** is shown with a RAC software agent **302** and a basic input output system (BIOS) **304**, and one or more device drivers **316** that are each associated with one of multiple hardware devices **306**. The hardware devices **306** may be any IHS configurable device, such as a graphics processing card, a network interface card, an input/output (I/O) (e.g., USB) expansion port card, a Wi-Fi interface card, a storage controller card, and the like. For example, the hardware devices **306** may be a hardware accelerator **250**, a storage drive **235a**, an I/O port **215**, a network controller **240**, such as described herein above with reference to FIG. 2. Some, most, or all hardware devices **306** communicate with the IHS **200** via system bus **320**, which in one embodiment, may include a Peripheral Component Interconnect Express (PCIe) bus.

(51) The RAC **230** may communicate with each of multiple hardware devices **306** via an i2c connection **308** such as described above. In other embodiments, the RAC **230** may communicate with each of the hardware devices **306** using other communication protocols, such as i3c SENSEWIRE connections, or serial peripheral interface (SPI) based connections. Additionally, the RAC **230** may also communicate with each of the hardware devices **306** through the system bus **320** using a Management Component Transport Protocol (MCTP) PCIe vendor defined message (VDM) channel **310**.

(52) The RAC **230** may also communicate with the IHS **200** through the PCIe VDM channel **310**, a Keyboard Controller Style (KCS) channel **312**, and/or a USB NIC channel **314**. The KCS channel **312** is provided as part of an intelligent platform management interface (IPMI) specification. In general, the IPMI specification defines several system interfaces for local access from the RAC software agent **302** and BIOS **304** to the RAC **230** of which the KCS protocol provides an ideal connection that does not require authentication for its operation.

(53) The RAC software agent **302** is a lightweight software service that is executed on the host IHS **200** to integrate certain operating system (OS) features with the RAC **230**. The RAC software agent **302** provides OS-related information to the RAC **230**, and may add capabilities such as LC log event replication into the OS log, WMI support (including storage), iDRAC SNMP alerts via OS, iDRAC hard reset and remote full Power Cycle. For example, the RAC software agent **302** may be an iDRAC Service Module (iSM) that is configured to operate with the integrated Dell remote access controller (iDRAC), which are both provided by DELL TECHNOLOGIES.

(54) The IHS **200** may receive a firmware update **322** that is to be installed on multiple hardware devices **306**. The firmware update **322** may include a verification module **324** that includes information for verifying the firmware update **322** following its installation on a hardware device **306**. If the IHS **200** is PLDM compliant, it can take advantage of updating firmware in parallel (e.g. concurrently) to all hardware devices **306** without rebooting the servers. This feature, however, may present a problem for IHSs **200** that have multiple hardware devices **306**. For example, if the firmware update **322** exhibits problems after installation, the firmware update **322** for each of the multiple hardware devices **306** needs to be rolled back, thus yielding excessive downtime and customer dissatisfaction. Embodiments of the present disclosure provide a solution to this problem by performing a firmware update on a single IHS, calculate a health score for that IHS, and using the calculated health score, determine whether to perform the firmware update on the other hardware devices in the IHS.

(55) FIGS. **4A** and **4B** illustrate a health score-based firmware updating method **400** depicting how the hardware devices **306a-n** of the IHS **200** may receive a firmware update according to one embodiment of the present disclosure. In one embodiment, the health score-based firmware updating method **400** may be performed in whole, or in part, by the RAC **230** described herein above. Within this example method **400**, all of the hardware devices **306a-n** are of the same type. That is, each of the hardware devices **306a-n** are configured to be updated using the same firmware update **322**. Initially, a firmware update **322** (e.g., a new software package or an updated version of an existing software package) is promoted or made available by a provider of the hardware device **306** that the firmware update **322** package supports.

(56) Initially at step **404**, a user communicates with the RAC **230** via Remote Management Interface **101** to initiate a firmware update to the relevant hardware devices **306a-n** of the IHS **200**. When the RAC **230** receives the firmware update **322**, it parses the verification module **324** to obtain one or more verification criteria at step **406**, and generates a verification suite at step **408**. The criteria may be any type that can be used to verify that the hardware devices **306a-n** function properly with the firmware update **322**. Examples of such criteria may include, for example, a particular sideband interface that the hardware devices **306a-n** support, any new commands and/or requests that are supported by the hardware devices, a list of modified commands and/or requests added to the firmware, a list of mandatory commands that the hardware devices are to support, any expected results for the commands and/or requests issued to the hardware devices, and an expected performance for certain CPU intensive operations (e.g. log retrieval).

(57) Steps **410-420** describe actions that may be performed to verify one or more test commands (e.g., new commands/requests or modified commands/requests). At step **410**, the RAC **230** begins a firmware update process for one of the hardware devices **306a-n**. Whereas the user via Remote Management Interface **101** may have requested that all of the hardware devices **306a-n** receive the same firmware update **322**, the RAC **230** commences the firmware update on only one hardware device **306a** for reasons that will be explained in detail herein below. When the firmware update process is completed, the RAC **230** will activate the one device **306a** at step **412** in which the hardware device **306a** responds by sending an activation complete message back to the RAC **230** at step **414**.

(58) At step **416**, the RAC **230** issues a test command to the hardware device **306a** to verify one criterion from among the criteria that was parsed and generated previously at steps **406** and **408**. Thereafter at step **418**, the RAC **230** receives a response to the test command. The RAC **230** performs a similar verification steps for the other criteria. In one embodiment, certain verification criteria tests may be conducted sequentially, while in other embodiments, certain other verification criteria tests may be conducted in parallel. The RAC **230** may then calculate a test score based upon a cumulative results of the test commands at step **420**. In some embodiments, the results of certain test commands may be weighted according to their relative importance to the overall health score of the firmware update **322**.

(59) Steps **422-426** describe actions that may be performed to verify the operation of the hardware device **306a** based upon information stored in a syslog of the IHS **200**. At step **422**, the RAC **230** fetches a syslog from the IHS **200**, and at step **424**, it receives the requested syslog from the IHS **200**. In general, the syslog may store information about any of the test commands that did, or did not, pass the tests. The RAC **230** may use such information to calculate another device test score based upon the information stored in the syslog. At step **426**, the RAC **230** generates a device test score based on the syslog.

(60) Steps **428-434** describe actions that may be performed to verify in-band communications (via RAC software agent **302**) between the RAC **230** and the hardware device **306a**. Nevertheless, it should be appreciated that steps **428-434** need not be performed, for example, if the hardware device **306a** does not need or use any in-band communication operations. At step **428**, the RAC **230** issues an in-band verification routine to the RAC software agent **302** which in turn, forwards it to the IHS **200** at step **430**. The in-band verification routine may include any type of suitable logic for verifying the integrity of the in-band communication link between the RAC **230** and hardware device **306a**. For example, the in-band verification routine may be configured to send arbitrary placeholder text (e.g., dummy data) that is compared with a known copy of the placeholder text to verify proper operation of the in-band communication link. At step **432**, the RAC software agent **302** receives the result of the routine, and forwards it onto the RAC **230** at step **434**.

(61) At step **436**, the RAC **230** calculates a cumulative score based on the score calculated for the test commands, syslog, and in-band verification routine. In other embodiments, the RAC **230** may calculate the cumulative score based on a portion of the aforescribed interim scores. In one example, the cumulative score may be based only on the test commands, while in another example, the cumulative score may be based on the test commands and the in-band verification routine.

(62) In one embodiment, the RAC **230** may display the cumulative score for view by the user via the Remote Management Interface **101**, and receive user input for determining whether to install the firmware update **322** on the other hardware devices **306b-n**. The RAC **230** may also receive user input for determining whether to revert (e.g., roll back) the firmware update **322** from the first hardware device **306a**. For example, if the score is poor, the RAC **230** may issue a request to revert the firmware update **322** in the first hardware device **306a** at step **438** in which the hardware device **306a** responds with a revert success message at step **440**.

(63) If, however, the score is good, the RAC **230** may trigger an update to the other hardware devices **306b-n** at step **442** using the firmware update **322** in which the hardware devices **306b-n** respond with respective update success messages at step **444**. When all of the other hardware devices **306b-n** have been successfully updated with the firmware update **322**, the RAC **230** send a firmware update success message to the Remote Management Interface **101** at step **446**.

(64) The aforescribed method **400** can be performed each time a firmware update **322** is to be installed on multiple hardware devices **306a-n** of the IHS **200**. Nevertheless, when use of the health score-based firmware updating method **500** is no longer needed or desired, the method **500** ends.

(65) Although FIG. **4** describes one example of a process that may be performed to update the hardware devices **306a-n** of the same type in an IHS **200**, the features of the disclosed process may be embodied in other specific forms without deviating from the spirit and scope of the present disclosure. For example, certain steps of the disclosed processes may be performed sequentially, or alternatively, they may be performed concurrently. As another example, the method **400** may perform additional, fewer, or different operations than those operations as described in the present example. As yet another example, the steps of the processes described herein may be performed by a system other than the RAC **230**, such as by a cloud service existing in the cloud network that communicates remotely with the IHS **200**.

(66) It should be understood that various operations described herein may be implemented in software executed by logic or processing circuitry, hardware, or a combination thereof. The order in which each operation of a given method is performed may be changed, and various operations may

be added, reordered, combined, omitted, modified etc. It is intended that the invention(s) described herein embrace all such modifications and changes and, accordingly, the above description should be regarded in an illustrative rather than a restrictive sense.

(67) Although the invention(s) is/are described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention(s), as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of the present invention(s). Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

(68) Unless stated otherwise, terms such as “first” and “second” are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements. The terms “coupled” or “operably coupled” are defined as connected, although not necessarily directly, and not necessarily mechanically. The terms “a” and “an” are defined as one or more unless stated otherwise. The terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has” and “having”), “include” (and any form of include, such as “includes” and “including”) and “contain” (and any form of contain, such as “contains” and “containing”) are open-ended linking verbs. As a result, a system, device, or apparatus that “comprises,” “has,” “includes” or “contains” one or more elements possesses those one or more elements but is not limited to possessing only those one or more elements. Similarly, a method or process that “comprises,” “has,” “includes” or “contains” one or more operations possesses those one or more operations but is not limited to possessing only those one or more operations.

Claims

1. An Information Handling System (IHS) comprising: at least one processor; and at least one memory coupled to the at least one processor, the at least one memory having program instructions stored thereon that, upon execution by the at least one processor, cause the IHS to: receive a request to perform a firmware update to a plurality of devices; identify one or more new or modified commands available in the requested firmware update; update a first device of the plurality of devices with the requested firmware update; issue each of the one or more new or modified commands to the first device of the plurality of devices; verify whether a response to the one or more new or modified commands was successful; calculate a health score for the first device of the plurality of devices based at least in part on the successful response of the one or more new or modified commands, as a result of the first device of the plurality of devices being updated; determine whether to update other devices of the plurality of devices with the requested firmware update based at least in part on the calculated health score for the first device of the plurality of devices; and in response to the determining, update the other devices of the plurality of devices with the requested firmware update based at least in part on the calculated health score for the first device of the plurality of devices.

2. The IHS of claim 1, wherein the program instructions, upon execution by the at least one processor, further cause the IHS to: fetch a syslog from the IHS, wherein the syslog comprises information associated with one or more errors reported about the first device of the plurality of devices after being activated with the requested firmware update; and further calculate the health score for the first device of the plurality of devices based at least in part on a relevance of the one or more reported errors to the one or more new or modified commands.

3. The IHS of claim 1, wherein the program instructions, upon execution by the at least one processor, further cause the IHS to: communicate with the first device of the plurality of devices via an in-band communication link, wherein the in-band communication link is performed using a

Peripheral Component Interconnect Express (PCIe) protocol; and further calculate the health score for the first device of the plurality of devices based at least in part on any errors reported due to the communication with the first device of the plurality of devices via the in-band communication link.

4. The IHS of claim 3, wherein the program instructions are stored and executed on a Remote Access Controller (RAC) configured in the IHS.

5. The IHS of claim 4, wherein the program instructions, upon execution by the at least one processor, further cause the IHS to communicate with the first device of the plurality of devices via a RAC service module executed on the IHS.

6. The IHS of claim 1, wherein the program instructions, upon execution by the at least one processor, further cause the IHS to: when the calculated health score for the first device of the plurality of devices exceeds a specified threshold, update the other devices of the plurality of devices with the requested firmware update; and when the calculated health score for the first device of the plurality of devices does not meet the specified threshold, roll back the requested firmware update on the first device of the plurality of devices.

7. The IHS of claim 1, wherein the program instructions, upon execution by the at least one processor, further cause the IHS to: display the calculated health score for the first device of the plurality of devices for view by a user; and receive user input for determining whether to update the other devices of the plurality of devices with the requested firmware update.

8. The IHS of claim 1, wherein the requested firmware update comprises a Platform Level Data Model (PLDM)-based firmware update.

9. A health score-based firmware updating method comprising: receiving a request to perform a firmware update to a plurality of devices; identifying one or more new or modified commands available in the requested firmware update; updating a first device of the plurality of devices with the requested firmware update; issuing each of the one or more new or modified commands to the first device of the plurality of devices; verifying whether a response to the one or more new or modified commands was successful; calculating a health score for the first device of the plurality of devices based at least in part on the successful response of the one or more new or modified commands, as a result of the first device of the plurality of devices being updated; determining whether to update other devices of the plurality of devices with the requested firmware update based at least in part on the calculated health score for the first device of the plurality of devices; and in response to the determining, updating the other devices of the plurality of devices with the requested firmware update based at least in part on the calculated health score for the first device of the plurality of devices.

10. The health score-based firmware updating method of claim 9, further comprising: fetching a syslog from an Information Handling System (IHS), wherein the syslog comprises information associated with one or more errors reported about the first device of the plurality of devices after being activated with the requested firmware update; and further calculating the health score for the first device of the plurality of devices based at least in part on a relevance of the one or more reported errors to the one or more new or modified commands.

11. The health score-based firmware updating method of claim 9, further comprising: communicating with the first device of the plurality of devices via an in-band communication link, wherein the in-band communication link is performed using a Peripheral Component Interconnect Express (PCIe) protocol; and further calculating the health score for the first device of the plurality of devices based at least in part on any errors reported due to the communication with the first device of the plurality of devices via the in-band communication link.

12. The health score-based firmware updating method of claim 11, further comprising communicating with the first device of the plurality of devices via a Remote Access Controller (RAC) service module executed on an Information Handling System (IHS).

13. The health score-based firmware updating method of claim 9, further comprising: when the calculated health score for the first device of the plurality of devices exceeds a specified threshold,

updating the other devices of the plurality of devices with the requested firmware update; and when the calculated health score for the first device of the plurality of devices does not meet the specified threshold, rolling back the requested firmware update on the first device of the plurality of devices.

14. The health score-based firmware updating method of claim 9, further comprising: displaying the calculated health score for the first device of the plurality of devices for view by a user; and receiving user input for determining whether to update the other devices of the plurality of devices with the requested firmware update.

15. A memory storage device having program instructions stored thereon that, upon execution by one or more processors of a client Information Handling System (IHS), cause the client IHS to: receive a request to perform a firmware update to a plurality of devices; identify one or more new or modified commands available in the requested firmware update; update a first device of the plurality of devices with the requested firmware update; issue each of the one or more new or modified commands to the first device of the plurality of devices; verify whether a response to the one or more new or modified commands was successful; calculate a health score for the first device of the plurality of devices based at least in part on the successful response of the one or more new or modified commands, as a result of the first device of the plurality of devices being updated; determine whether to update other devices of the plurality of devices with the requested firmware update based at least in part on the calculated health score for the first device of the plurality of devices; and in response to the determining, update the other devices of the plurality of devices with the requested firmware update based at least in part on the calculated health score for the first device of the plurality of devices.

16. The memory storage device of claim 15, wherein the program instructions, upon execution by the one or more processors, further cause the client IHS to: fetch a syslog from the client IHS, wherein the syslog comprises information associated with one or more errors reported about the first device of the plurality of devices after being activated with the requested firmware update; and further calculate the health score for the first device of the plurality of devices based at least in part on a relevance of the one or more reported errors to the one or more new or modified commands.

17. The memory storage device of claim 15, wherein the program instructions, upon execution by the one or more processors, further cause the client IHS to: communicate with the first device of the plurality of devices via an in-band communication link, wherein the in-band communication link is performed using a Peripheral Component Interconnect Express (PCIe) protocol; and further calculate the health score for the first device of the plurality of devices based at least in part on any errors reported due to the communication with the first device of the plurality of devices via the in-band communication link.
